Resilience Academy

SDI Training for ICT Professionals

KoboToolbox Overview

Mobile Data Collection for Fieldwork

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Day 3 | 1 Hour Lecture | crd.resilienceacademy.ac.tz

Resilience Academy

What is KoboToolbox?

A Free Tool for Collecting Data in the Field

- **Open-source** platform for mobile data collection
- Used by **600,000+** organisations worldwide
- Created at Harvard for humanitarian work in challenging environments
- Supported by **OCHA** (UN Office for Coordination of Humanitarian Affairs)

Design your form (web browser)
↓
Deploy to phones (one click)
↓
Collect in the field (offline capable)
↓
View & export data (web browser)

Key Advantage

Collect data on your phone **even without internet** – GPS coordinates, photos, survey answers – and

Real-World Example: Tanga Flood Mapping

Resilience Academy / GIZ Project

16 Resilience Academy alumni used KoboToolbox to collect flood data in Tanga, Tanzania.

What they collected:

- GPS coordinates of flooded areas
- Photos of flood damage
- Water depth estimates
- Infrastructure impact (roads, buildings)
- Timestamps for each observation

How they did it:

- Installed **KoboCollect** app on Android phones
- Walked through flood-affected neighborhoods
- Filled in survey forms at each location
- Phone recorded GPS automatically
- Uploaded data when back at the office

Result

Use Cases: Where KoboToolbox Fits in Government Work

Disaster Assessment

- Damage surveys after floods, storms
- Record locations of affected buildings
- Photo evidence for insurance / aid
- Real-time situation reports

Health Surveys

- Household health assessments
- Water quality sampling locations
- Vaccination coverage tracking
- Facility condition surveys

Infrastructure Audits

- Map condition of roads and bridges
- GPS-locate water points and wells
- Record status of public buildings
- Track maintenance needs

Environmental Monitoring

- Forest cover observation points
- Coastal erosion measurements
- Waste dump site mapping
- Mangrove restoration tracking

Cloud vs Self-Hosted: Why Run Your Own Server?

Option A: Use kobo.humanitarianresponse.info

- Free cloud service by OCHA
- No installation needed
- Sign up and start immediately

× Data stored outside Tanzania
× No control over server
× Subject to usage limits
× Depends on external availability

Option B: Self-Hosted (What We Will Do)

✓ **Data sovereignty** – data stays in your country
✓ **Government compliance** – meets local data policies
✓ **Full control** – you manage users and backups
✓ **No usage limits** – unlimited submissions
✓ **Custom branding** – your organisation's name

- Requires a server with Docker

KoboCollect App: Offline GPS Data Collection

What is KoboCollect?

A free Android app that lets field workers fill in survey forms on their phones – **no internet needed** during data collection.

How it works:

1. Open **KoboCollect** on your Android phone
2. Tap **"Get Blank Form"** (needs internet once)
3. Go to the field – **no internet needed**
4. Fill in forms, phone captures GPS automatically
5. Take photos directly within the form
6. When back online, tap **"Send Finalized Form"**

KoboCollect

- Fill Blank Form
- Edit Saved Form
- Send Finalized Form
- Get Blank Form
- Delete Saved Form

Works offline
GPS auto-capture

The Complete Workflow: Form → Deploy → Collect → Analyse

1. Design
Build form in web browser

Drag-and-drop question builder

→

2. Deploy
Click "Deploy" button

Form becomes available on phones

→

3. Collect
Field workers use phones

GPS, photos, answers saved offline

→

4. Analyse
View data, export files

Maps, tables, CSV/GeoJSON export

What You Need

- A computer with a web browser (for steps 1, 2, 4)
- Android phones with KoboCollect (for step 3)
- Internet access at office (not needed in

No Programming Required

- Form design is visual (drag and drop)
- Deployment is one click
- Data export is done through menus
- Everything happens in a web browser

Live Demo: What You Will See

The instructor will now demonstrate KoboToolbox live on the projector.

Follow along on screen – you will do this yourself in the practical session.

Demo steps:

1. **Login page** – Open the KoboToolbox web interface in a browser
2. **Dashboard** – See the list of forms (projects)
3. **Open a deployed form** – The Tanga Flood Survey form
4. **View submissions** – See the data table with all collected entries
5. **Map view** – See GPS points plotted on a map
6. **Single submission** – Click one entry to see its details and photo
7. **Export menu** – Show the download options (CSV, Excel, GeoJSON)

What to Notice

- Every submission has a **GPS coordinate** – that is what makes this data *spatial*

Key Takeaways and What Comes Next

What We Learned

- KoboToolbox is a **free, open-source** tool for collecting data in the field
- It captures **GPS coordinates** automatically – making your data spatial
- The **KoboCollect app** works offline on Android phones
- Self-hosting keeps **data under your control**
- The workflow is simple: **Design** → **Deploy** → **Collect** → **Analyse**

Coming Up Today

S2 **Installation** – Set up KoboToolbox on your training server using Docker

S3 **Form Design** – Build the Tanga Flood Survey form step by step

S4 **Administration** – Manage users, export data, run backups

S5 **Integration** – Move your Kobo data into QGIS and the CRD